What do X-ray colours capture, and what does low-frequency variability add?

Black hole versus neutron star classification of X-ray binaries excluded from training, with RXTE and NICER

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Abstract

Classifiers that assign a black hole (BH) or neutron star (NS) to an X-ray binary from a single observation are usually summarised by one accuracy, which mixes three questions: which part of the data carries the signal, in which accretion states, and whether it holds for unseen sources. We address them with evaluations in which every test source is excluded from training, using RXTE/PCA and NICER/XTI observations of dynamically confirmed BHs and of bursting or pulsating NSs. For 31 RXTE sources (7 BHs, 456 observations), a logistic regression on two hardness ratios reached a source-level area under the ROC curve (AUC) of 0.923 (95% source-bootstrap interval 0.804–0.994). The full 5–25 keV spectrum, extra energy bands, 17 times more observations and a nested search over eight algorithms added no detectable improvement, which does not demonstrate equivalence. The colour-only observation-level AUC was 0.954 in observations with weak 0.1–4 Hz variability ("soft-like") but 0.646 in observations with strong variability ("hard-like"). Within hard-like observations, adding fractional rms in three bands between 0.016 and 4 Hz raised the observation-level AUC in RXTE (+0.264 [+0.048, +0.455]) and, with a random forest, in each NICER set. On 198 previously unanalysed NICER observations, processed with a frozen pipeline that included 3C50 background correction and classified with a random forest specified before download, the hard-like AUC rose from 0.680 to 0.864 (+0.184 [+0.049, +0.337]; 74 observations of 8 BHs and 19 NSs); the source-level gain was +0.125 [−0.053, +0.303]. These observations come from sources already in the sample, the analysis plans were written after earlier results, and only 16 confirmed BHs had usable timing data, so confirmation on new sources is still required. A lower characteristic variability frequency of BHs at matched colour appeared in every NICER set, but its significance depended on the background treatment.

Keywords: X-rays: binaries — accretion — stars: black holes — stars: neutron — methods: statistical **[journal keyword list to be matched to the target journal]**

1 Introduction

The nature of the compact object in an X-ray binary is established securely in only two ways. A dynamical mass above the maximum NS mass, usually measured from optical or infrared spectroscopy of the donor star in quiescence, identifies a BH (Corral-Santana et al., 2016; Marcel

et al., 2026). Thermonuclear (type I) X-ray bursts or coherent pulsations identify an NS (Galloway et al., 2008, 2020). Many X-ray binaries, including most newly discovered transients during their first outburst, have neither. In practice the compact object is often inferred from X-ray phenomenology: the shape of the spectrum, its evolution through accretion states, and the variability (Remillard and McClintock, 2006).

What is known physically. An NS has a surface, whereas a BH has an event horizon. Emission from the NS surface or boundary layer changes the X-ray spectrum most clearly when the source is bright and soft. Done and Gierliński (2003) compared RXTE colour–colour diagrams of atoll sources and BH binaries and attributed the systematic differences between their tracks to the presence of an NS surface. In the hard state, BH and NS spectra are more alike in the 2–20 keV range. Burke et al. (2017) found that their differences lie mainly in Comptonization parameters (electron temperature and Compton y), which require broad-band spectral fitting. Variability has long been used as a discriminant. Sunyaev and Revnivtsev (2000) proposed that significant power above a few hundred Hz in the low/hard state indicates an NS. At lower frequencies, broad-band noise and quasi-periodic features follow similar frequency correlations in BH and low-luminosity NS systems (Wijnands and van der Klis, 1999). However, the highest characteristic frequencies of BHs lie several times below those of NSs (Klein-Wolt and van der Klis, 2008). Power colours show that NS low-mass X-ray binaries move through tracks that closely resemble those of BHs (Heil et al., 2015; Gardenier and Uttley, 2018). Low-frequency variability is therefore known to be similar in BHs and NSs in many respects. It is not obvious whether it contains information that complements the spectral colours.

What machine-learning studies have done. Pattnaik et al. (2021) trained a random forest on RXTE/PCA 5–25 keV spectra of 61 low-mass X-ray binaries (about 15 000 observations). They evaluated it with whole sources held out of training and reported an average accuracy of 87 ± 13 per cent. They also noted that misclassifications are more frequent in hard and intermediate states and at low signal-to-noise ratio. de Beurs et al. (2022) compared Gaussian-process, k-nearest-neighbour and support-vector classifiers on MAXI colour–colour–intensity diagrams, separating BHs, non-pulsing NSs and pulsars, again with sources held out. Garg et al. (2026) trained neural networks on the RXTE spectra of the Pattnaik et al. sample and reported accuracies of about 90–94 per cent. Their training and test sets were split over observations rather than over sources (as recorded in the project's reading of that paper; see Appendix D). To interpret the networks, they fitted each spectrum with a disc-plus-power-law model. They identified the blackbody temperature, photon index, flux ratio, reduced χ^2 and the variance of the spectral data as likely drivers. This "variance" is a property of the spectral data. It is not a Fourier variability measure. Source-held-out evaluation, the difficulty of hard states, and the diagnostic value of timing are therefore all established.

What is not yet answered. Three questions remain open for sources excluded from training. First, how much of the performance of spectral classifiers can be reproduced by two hardness ratios, and does anything else in the spectrum add detectable information? Second, how large is the state dependence once the uncertainty respects the clustering of observations within sources, rather than being summarised by a pooled accuracy? Third, within hard-like observations, where colours overlap, does low-frequency variability add information beyond the colours? If it does, does the gain survive a change of instrument, a change of background treatment, and new observations analysed with a pipeline fixed in advance?

This paper. We integrate a sequence of analyses carried out in one repository (Appendix A) and organise them around these questions. Section 2 describes the RXTE and NICER samples, and Section 3 the features, models and evaluation. Section 4 presents, in turn, the colour baseline and spectral ablations, the state dependence, the hard-like timing gain, and the background and new-observation tests; a secondary characteristic-frequency result is summarised in Section 4.5. Section 5 separates what has been learned from what remains unproven. We do not propose a new classifier that competes with published accuracies. Those accuracies were obtained with different samples, labels, metrics and data splits, and are not comparable with the numbers reported here.

2 Data and sample construction

2.1 Labels

BHs were restricted to dynamically confirmed systems: BlackCAT (Corral-Santana et al., 2016) for the first RXTE sample, and Table 1 of Marcel et al. (2026) for later samples. NSs were type-I bursters listed by Galloway et al. (2008) or in MINBAR (Galloway et al., 2020); the external RXTE set also included accretion-powered pulsars. BH candidates were never used as training labels. Across both instruments the union of usable sources is 86, of which 16 are dynamically confirmed BHs. These are nearly all of the 25 dynamically confirmed BHs of Marcel et al. (2026) that have usable RXTE or NICER timing data; the others are mostly in quiescence, pre-date RXTE, or lack suitable X-ray timing data (Section 5.2).

2.2 RXTE/PCA samples

The *development sample* contains every source that satisfied the label rule, had an RXTE catalogue file, and had at least 10 eligible pointings in PCA gain epoch 5 (2000 May 13 to the end of the mission). Eligible pointings had a catalogue exposure of at least 1000 s and pointed within 0.1° of the source. Up to 15 pointings per source were drawn, one at random from each of 15 equal-count time bins. After quality cuts (spectral exposure ≥ 1000 s, net 5–25 keV signal-to-noise ≥ 10, dead-time fraction $\leq 10\%$), the sample has 31 sources (7 BHs, 24 NSs) and 456 observations. Spectra are HEASARC Standard Products (Standard-2 source spectra with PCABACKEST background) on a common 45-bin 5–25 keV grid (Jahoda et al., 2006), with dead time corrected from Standard-1 data. Low-frequency timing uses Standard-1 light curves with 0.125 s resolution. A later check against Galloway et al. (2008) found 13 more bursters with eligible data that the source rule should have included. The development sample was not changed, and this is listed as a limitation.

The *external RXTE set* contains sources that were never used to train the development-sample models: Cyg X-1, LMC X-1, LMC X-3, GS 1354−64, SS 433, V4641 Sgr and XTE J1859+226 (BHs), and 19 NSs, including six accreting millisecond pulsars. It has 379 observations; a separate stress set of four slowly rotating pulsars is excluded here. These sources span all gain epochs, so their colours were divided by the colours of a $\Gamma = 2$ power law folded through each observation's response. Fifteen of these sources had been examined in earlier analyses with other models. Observations containing type-I bursts were identified with an automatic light-curve rule and with MINBAR. In the development sample, 58 of 456 observations overlapped a catalogued burst. They were kept in the main RXTE analyses, and excluding them, as a sensitivity analysis, did not change the colour-baseline conclusions (Appendix B). Only the rms used for the state classes (Section 3.3) excluded the burst intervals.

2.3 NICER/XTI observation sets

NICER sources were the dynamically confirmed BHs of Marcel et al. (2026) and the MINBAR NS entries with at least five eligible observations (target within 0.05°, exposure ≥ 1000 s). For each source, observations were sorted in time and divided into eight bins, and up to three candidates per bin were tried. Cleaned event files were read from the HEASARC archive. An observation was accepted if it had at least three gap-free 128-s segments with at least two active measurement power units (MPUs) and a mean 2–10 keV rate of at least 20 counts s^{-1}. Segments overlapping automatically detected bursts were removed. Three sets of observations of the *same* source list were analysed. They are disjoint by ObsID, and this was checked explicitly:

- **N1** (development): 368 accepted observations of 63 sources (9 BHs, 54 NSs). It was analysed first using the beginning of each event file, and then using the full files (366 usable).
- **N2**: 318 observations never downloaded before. All but 12 belong to N1 sources; the 12 come from sources that were never used before, all NSs.
- **N3**: 238 further observations, disjoint from N1 and N2 (37 sources: 8 BHs, 29 NSs), of which 198 (35 sources) survived background screening (Section 2.4).

N2 and N3 are new *observations*, not new *sources*. Several BHs, such as MAXI J1820+070 and Swift J1727.8−1613, had a single bright outburst during the NICER era. Their N1, N2 and N3 observations therefore sample the same outburst at different times.

2.4 Background treatment

For RXTE, the Standard Products background was subtracted from the spectra, and timing amplitudes were normalised to the source count rate. For NICER, N1 and N2 were first analysed without background subtraction; a 12–15 keV background proxy was used only in sensitivity analyses. The combined N1+N2 data were later re-analysed, and N3 was processed, with the 3C50 empirical background model (Remillard et al., 2022). It was applied to every observation with a 2–10 keV rate below 500 counts s^{-1}. With f the background fraction in a band:

- fractional rms amplitudes were multiplied by $1/(1-f)$;
- colours were recomputed from background-corrected band counts;
- the state class was reassigned from the corrected rms;
- observations with $f > 0.20$ in 2–10 keV, or with a failed 3C50 run, were excluded.

In N1+N2, 3C50 succeeded for 362 of 503 faint observations. Of the 141 failures, 139 were files processed with an early gain calibration that the 3C50 tool does not accept. The median f was 0.008, nine observations had $f > 0.20$, and 534 of 684 observations were retained. In N3, 109 of 148 faint observations succeeded (38 of the 39 failures had the old gain calibration), the median f was 0.006, and 198 of 238 observations were retained. Three caveats apply:

- The excluded observations are not a random subset: they are mainly faint observations from 2017–2018.
- 3C50 removes high-background intervals from its own good-time intervals (GTIs), whereas the timing features use all GTIs, so f may underestimate the background in the timing data. For N3, this was checked by recomputing the faint observations' features within the 3C50 GTIs.

Table 1: Samples used in this paper. Hard-like counts refer to the rms-defined class of Section 3.3; numbers in parentheses are BH/NS sources. "Dev." = development data used to form hypotheses; "unseen" = never downloaded or analysed before the plan for that test was committed.

Sample	Instr.	Obs.	Sources (BH/NS)	Hard-like obs (BH/NS src)	Background	Evaluation
Development	PCA	456	31 (7/24)	180 (7/17)	PCABACKEST	LOSO
External	PCA	379	26 (7/19)	99 (5/15)	PCABACKEST	frozen models
N1	XTI	366	63 (9/54)	144 (7/32)	none	LOSO
N2 (unseen)	XTI	318	60 (9/51)	143 (9/33)[a]	none	frozen out-of-source
N1+N2	XTI	534[b]	58 (9/49)	202 (8/34)	3C50	LOSO
N3 (unseen)	XTI	198[c]	35 (8/27)	74 (8/19)	3C50	frozen out-of-source

[a]Observations of N1 sources only (the primary N2 test). [b]After 3C50 correction and screening of 684 observations. [c]Of 238 accepted. N1 counts are for full files; the N1 file-prefix analysis had 368 observations, 142 of them hard-like.

- Time variability of the background is not modelled, and the 3C50 libraries predate the 2023 optical light leak that affects Swift J1727.8−1613.

Table 1 summarises the samples, and Figure 1 shows how they were used.

3 Features, models and evaluation

3.1 Colours and spectral representations

The baseline representation H consists of two hardness ratios of net count rates in detector space. For RXTE they are $c_1 = F(7\text{–}10)/F(5\text{–}7\,\mathrm{keV})$ and $c_2 = F(16\text{–}25)/F(10\text{–}16\,\mathrm{keV})$, per PCU. For NICER they are $c_1 = C(4\text{–}6)/C(2\text{–}4\,\mathrm{keV})$ and $c_2 = C(6\text{–}10)/C(4\text{–}6\,\mathrm{keV})$. These are instrument-specific quantities, and no model was transferred between instruments. The RXTE ablations used five further representations:

- the 45-bin 5–25 keV spectral shape normalised by total flux (B);
- an intensity-preserving asinh transform of the spectrum (A);
- the colours plus log flux (HI);
- the concatenation H+B;
- the same quantities on a 3–25 keV grid (H3, B3).

For a subset of 396 observations, HEXTE 25–40 and 40–60 keV colours were also available.

3.2 Low-frequency variability features

Each light curve was divided into 128-s segments. In each segment we computed cross-spectra between independent detector groups (PCUs for RXTE, MPUs for NICER), averaged over all pairs. Because the Poisson and dead-time noise of different detectors is uncorrelated, the real part of the cross-spectrum has zero expected white-noise level. Fractional variance in a band was estimated as the ratio of the summed cross-power to the summed products of mean rates (a ratio of means over segments). The feature is the signed square root, $T_k = \mathrm{sign}(R_k)\sqrt{|R_k|}$, in three bands: $T_1 = 0.016\text{–}0.094$ Hz, $T_2 = 0.10\text{–}1.0$ Hz and $T_3 = 1.0\text{–}4.0$ Hz. The bands are the same for both instruments, but the energy coverage differs: RXTE Standard-1 records all PCA channels without energy resolution, whereas NICER features use 2–10 keV events. The model with these three features added to the colours is called H+T. The models never used the total rms itself as an input.

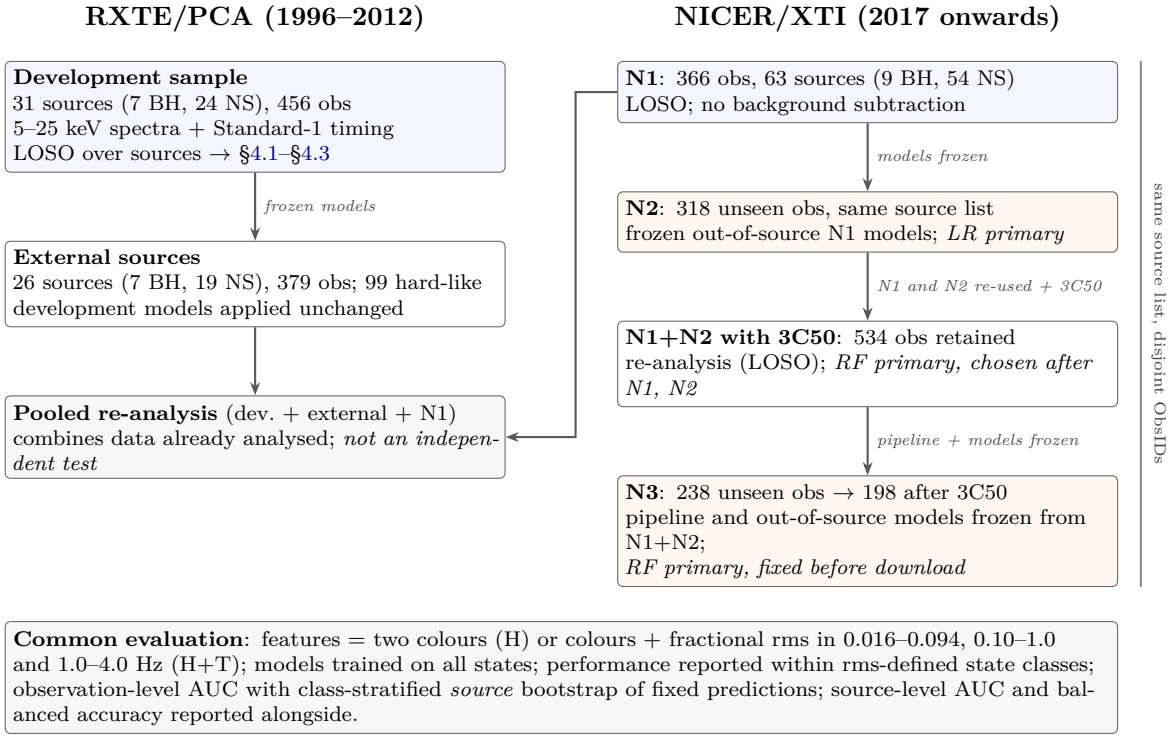


Figure 1: How the samples were used. Arrows show which models or pipelines were carried forward unchanged. Shaded orange boxes are the two tests on observations that had not been downloaded when the plan for that test was committed. The pooled re-analysis and the N1+N2 3C50 re-analysis reuse data and are robustness analyses, not independent replications. All NICER sets draw on one source list, so none of them tests generalisation to new sources.

3.3 Operational state classes

Each observation was assigned a state class from its fractional rms r, defined as follows:

- r is measured in 0.1–4 Hz for RXTE (limited by Standard-1) and in 0.1–10 Hz for NICER;
- hard-like: $r > 0.10$; soft-like: $r < 0.075$; intermediate otherwise;
- the thresholds are the rms criteria of Remillard and McClintock (2006), used without their spectral and QPO criteria.

These classes are operational. They are not full spectral-state assignments. Because the RXTE band stops at 4 Hz, its r is lower than the 0.1–10 Hz value, which makes "hard-like" conservative for RXTE. The classes were checked against expectations: XTE J1118+480 was hard-like in 15 of 15 RXTE observations, and the Z sources Cyg X-2 and GX 17+2 were soft-like in 27 of 28 RXTE and 15 of 16 NICER observations. Labels were not used to define the classes. All models were trained on all states, and the classes were used only to stratify the evaluation. The stratifying variable is itself a variability amplitude that overlaps T_2 and T_3 in frequency; Section 5.2 discusses what this implies.

3.4 Models

We used two classifiers from scikit-learn (Pedregosa et al., 2011):

- a logistic regression (LR) with an L_2 penalty ($C = 1$), class-balanced weights and standardised inputs;

- a random forest (RF) with 500 trees, $\sqrt{p}$ features per split, a minimum of two samples per leaf and class-balanced weights.

Hyperparameters were fixed in advance and not tuned. A nested search over eight algorithms is reported only as an ablation.

3.5 Evaluation designs and metrics

Three designs were used.

1. *Leave-one-source-out (LOSO)* within a sample: all observations of a source are scored by a model trained on all other sources.
2. *Frozen out-of-source prediction* on unseen observations (N2, N3): the observations of source s are scored by a model trained on the reference data (N1, or the corrected N1+N2) with s removed. Neither the features, the state thresholds, the background treatment nor the models were changed after the new data were seen.
3. *Frozen external prediction* (RXTE external set): development-sample models applied unchanged.

We distinguish four metrics:

- *observation-level AUC*: the probability that a randomly chosen BH observation receives a higher BH score than a randomly chosen NS observation, pooling all observations in a stratum, so sources with more observations weigh more;
- *observation-level balanced accuracy*: the mean of the BH and NS recalls at the fixed threshold 0.5;
- *source-level AUC*: computed from source scores, each defined as the mean of the observation scores of that source;
- *source-level balanced accuracy*: the same recalls computed on source scores at threshold 0.5.

The two source-level metrics weight each source equally. An AUC measures ranking, not accuracy: an AUC of 0.864 does not mean that 86.4% of observations are classified correctly.

Intervals are 2.5–97.5 percentiles over 2000 class-stratified bootstrap draws of *sources* (BH and NS sources resampled separately with replacement, seed 42). In each draw the metric was recomputed on the fixed out-of-sample predictions; models were not refitted. For stratified metrics, the draws are over all sources of a sample, and the stratum is applied within each draw. Differences between models are paired within draws. A difference was declared "detected" when its interval excluded zero, and "not detected" otherwise. These intervals describe source-sampling variability given fixed predictions. They do not include training variability, measurement error in the features, label uncertainty, or the multiplicity of analyses across the project.

3.6 Analysis plans and the status of each result

Each analysis round had a plan document committed to the version-controlled repository before the data for that round were downloaded (Table 2). We call these *time-stamped, internally pre-specified analysis plans.* Each plan was written after the results of all earlier rounds were known, and plans were drafted and approved within the project without external registration or review. Commit timestamps establish the order of events within the repository; they are not an external registry. Every result below is labelled as one of three types:

Table 2: Provenance of the primary hard-like timing tests. "Plan commit" is the repository commit that added the plan document (time UTC+8, 2026). Results are hard-like observation-level AUC differences, H+T − H, with 95% class-stratified source-bootstrap intervals. All plans were written after the results of earlier rounds.

Test (data status)	Plan commit (UTC+8)	Primary model	ΔAUC (95% interval)	Verdict
RXTE development (development data)	`a481fa0`, Oct 5 21:14	LR	+0.264 [+0.048, +0.455]	detected
RXTE external (new sources)	`a481fa0`	LR, frozen	+0.064 [−0.185, +0.367]	not detected
NICER N1 (new instrument)	`ee4b7db`, Oct 5 23:07	LR	+0.112 [−0.004, +0.229]	not detected
Pooled (re-use of rows 1–3)	`aeedac7`, Oct 6 00:10	LR	+0.145 [+0.028, +0.257]	detected[a]
NICER N2 (unseen observations)	`6fea016`, Oct 6 01:43	LR, frozen	+0.066 [−0.095, +0.180]	not detected
N1+N2 with 3C50 (re-use)	`2126e9d`, Oct 6 03:13	RF[b]	+0.174 [+0.071, +0.300]	detected[a]
NICER N3 (unseen observations)	`42fe1d2`, Oct 6 04:53	RF, frozen[c]	+0.184 [+0.049, +0.337]	detected

[a]Not an independent test: the data were analysed in earlier rounds. [b]RF chosen as primary after RF gains had been seen in N1 and N2. [c]RF, the 3C50 pipeline and the out-of-source models were fixed before the N3 data were downloaded; the N3 data commit is `6867de0` (05:35) and the results commit `3d86930` (06:46).

- *primary*: the one or two comparisons declared in that round's plan;
- *descriptive*: other comparisons declared in the plan;
- *post hoc*: comparisons added after the results were seen.

The model class used for the primary hard-like test changed during the project. LR was primary through N2. RF became primary for the N1+N2 re-analysis after its gains had been seen in earlier rounds. RF was fixed before download only for N3. No correction for multiple comparisons was applied across rounds.

4 Results

4.1 Two colours as a spectral baseline

On the development sample, LR on the two colours (H-LR) reached:

- source-level AUC 0.923 [0.804, 0.994];
- source-level balanced accuracy 0.795 [0.589, 0.958], with 26 of 31 sources correct (5 of 7 BHs, 21 of 24 NSs);
- observation-level balanced accuracy 0.776 [0.633, 0.893].

RF on the same colours gave a source-level AUC of 0.905 [0.750, 1.000] (Figure 2a). The misclassified sources under H-LR were the BHs GRS 1915+105 and XTE J1118+480 and the NSs HETE J1900.1−2455, KS 1731−260 and SAX J1808.4−3658.

Adding more spectral information produced no detectable improvement (Figure 2b). The source-level AUC differences relative to H were:

- full 5–25 keV shape (H+B, primary): −0.048 [−0.179, +0.048] for LR and −0.012 [−0.077, +0.048] for RF;
- colours including 3–5 keV: −0.018 [−0.071, 0.000] for LR and −0.054 [−0.161, +0.006] for RF;

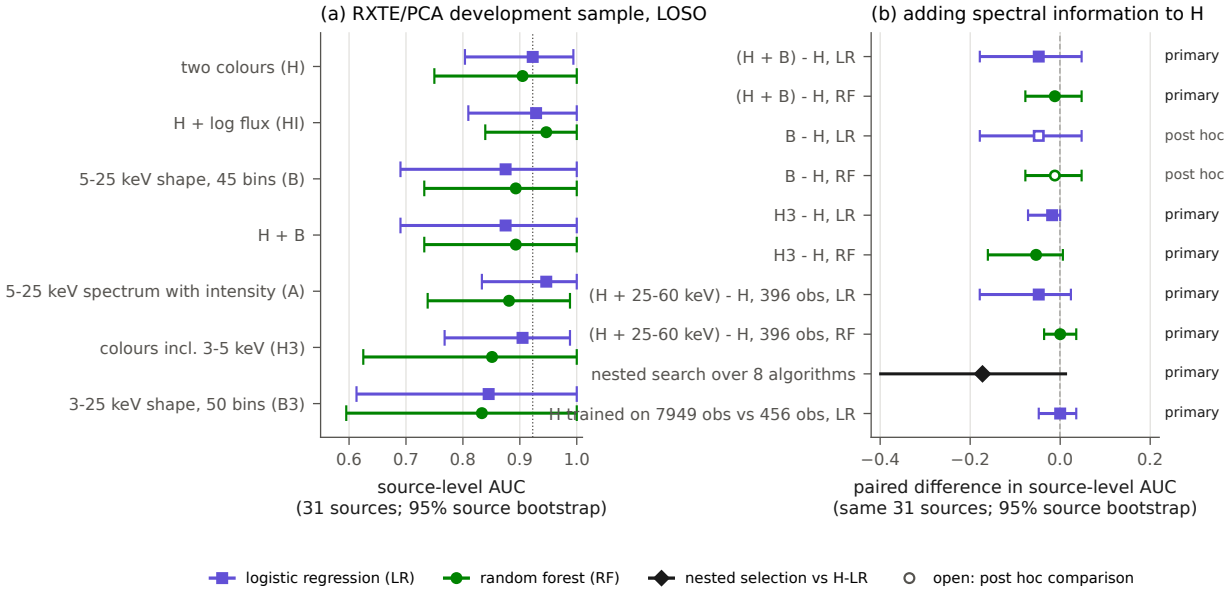


Figure 2: Colour baseline and spectral ablations on the RXTE development sample (31 sources, 456 observations, LOSO). (a) Source-level AUC (source score = mean observation score) for LR (violet squares) and RF (green circles); the dotted line marks H-LR. (b) Paired differences in source-level AUC relative to two colours. "Primary" and "post hoc" follow the time-stamped plans: the B−H comparison was added after the first results. The HEXTE rows use the 396 observations with HEXTE data; the nested search (black diamond) compares an automatically selected model with H-LR; the last row compares H-LR trained on all 7949 eligible epoch-5 observations with H-LR trained on the 456 used here, on the same 31 sources. Intervals are 95% class-stratified source-bootstrap percentiles of fixed LOSO predictions; they do not include training variability. Source: `results/v3/v3a_bootstrap.csv`, `v3b_bootstrap.csv`, `results/v4b3/v4b3_bootstrap.csv`, `results/v4/v4a_bootstrap.csv`, `results/v4b1/v4b1_bootstrap.csv`.

- 25–60 keV HEXTE colours (396 observations): −0.048 [−0.179, +0.024] for LR and 0.000 [−0.036, +0.036] for RF;
- training H-LR on all 7949 eligible epoch-5 observations instead of 456: 0.000 [−0.048, +0.036];
- a nested search over eight algorithms and four representations, selected inside the training sources: −0.173 [−0.399, +0.012] relative to H-LR.

Splitting the same data by observation instead of by source inflated observation-level balanced accuracy by +0.04 to +0.16, depending on representation and model.

Within the resolution of this sample, the BH/NS information measurable in 5–25 keV PCA spectra is therefore carried by the two colours. The intervals still allow losses of up to about 0.18 and gains of up to about 0.05 in source-level AUC, so the results do not establish that the colours contain all of the information, or that the representations are equivalent.

4.2 State dependence of colour information

Figure 3a–d shows the colour–colour planes split by state class. In soft-like observations, BH and NS observations occupy largely separate regions in both instruments. In hard-like observations they overlap. In the RXTE development sample:

- 221 observations were soft-like (38 BH observations from 5 BHs; 183 NS observations from 22 NSs);
- 180 were hard-like (59 BH observations from all 7 BHs; 121 NS observations from 17 NSs);
- 55 were intermediate or could not be classified.

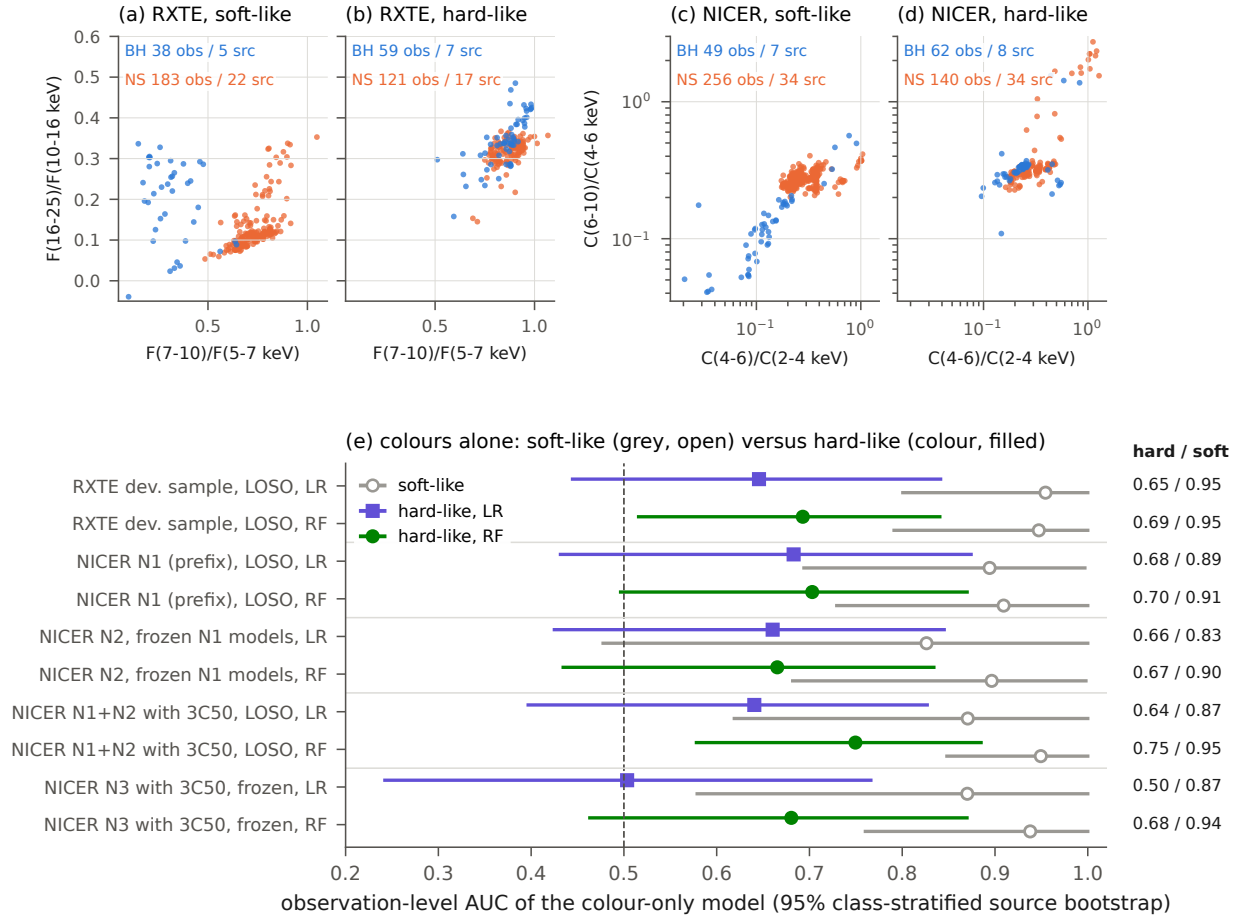


Figure 3: State dependence of the colour information. (a–d) Colour–colour planes for soft-like ($r < 0.075$) and hard-like ($r > 0.10$) observations, where r is the fractional rms in 0.1–4 Hz (RXTE) or 0.1–10 Hz (NICER). RXTE: development sample; NICER: N1+N2 after 3C50 correction and screening. Colours are detector-space count ratios and are not comparable between instruments; NICER axes are logarithmic. Intermediate observations are not shown. (e) Observation-level AUC of colour-only models in soft-like (grey, open) and hard-like (colour, filled) observations. Only the RXTE H-LR soft-minus-hard difference was a primary test; the NICER rows are descriptive. Intervals are 95% class-stratified source bootstrap of fixed out-of-sample predictions; numbers at right are the point estimates. Source: `results/v7a/states.csv`, `state_stratified_metrics.csv`, `results/v9a/v9a_metrics.csv`, `results/v11/v11a_metrics.csv`, `results/v12/v12b_gain.csv`, `results/v13/v13a_gain.csv`, `data/v12/processed/nicer_corrected_screened.csv`.

The H-LR observation-level AUC was 0.954 [0.801, 1.000] in soft-like observations and 0.646 [0.445, 0.841] in hard-like ones. The difference, +0.309 [+0.099, +0.510], was the primary test of that round. RF and the full spectral shape gave the same pattern, with soft-minus-hard differences of +0.254 for H-RF and +0.293 for B-LR, both with intervals excluding zero.

The same ordering holds in every NICER set (Figure 3e). Colour-only hard-like AUCs were 0.50–0.68 for LR and 0.67–0.75 for RF, against soft-like AUCs of 0.83–0.95. These NICER values are descriptive. Two features of the design matter for interpretation. Only five BHs contribute soft-like RXTE observations, so the soft-like AUC is uncertain. And the hard-like class contains all seven RXTE BHs, so the hard-like problem is the one that every BH source poses at some time.

4.3 Low-frequency variability within hard-like observations

Figure 4 collects every hard-like comparison of H+T with H using the same model class. Rows are grouped by sample and annotated with data status, background treatment, role, and when the model was fixed.

RXTE. In the development sample, adding T_1–T_3 raised the H+T-LR hard-like observation-level AUC from 0.646 to 0.910, a gain of +0.264 [+0.048, +0.455] (primary). The other comparisons in this sample were descriptive:

- with RF the gain was +0.230 [+0.074, +0.408];
- in soft-like observations, where the colours already separate the classes, the LR gain was +0.036 [0.000, +0.158];
- at source level within hard-like observations, the AUC rose from 0.748 to 0.933 (+0.185 [+0.008, +0.403]).

On the external RXTE sources, frozen development-sample models gave a hard-like gain of +0.064 [−0.185, +0.367] (primary, not detected; 99 observations of 5 BHs and 15 NSs). The colour-only baseline on these sources was already 0.817. Some sources moved in the right direction: GS 1354−64 rose from a mean BH score of 0.54 to 0.97. Others moved the wrong way: the NSs IGR J00291+5934 and EXO 1745−248 rose from 0.45 to 0.89 and from 0.32 to 0.73 respectively.

NICER N1. With LR the gain was +0.112 [−0.004, +0.229] (primary, not detected). With RF it was +0.236 [+0.104, +0.372] (descriptive). Using the full event files gave +0.107 [−0.010, +0.224] with LR and +0.219 [+0.100, +0.337] with RF. Adding the bands one at a time to the colours with LR gave +0.048 [−0.041, +0.140] for T_1, +0.143 [+0.028, +0.271] for T_2 and +0.115 [+0.030, +0.217] for T_3. In RXTE, by contrast, T_1 alone gave the largest single-band improvement (hard-like AUC 0.926, against 0.874 for T_2 and 0.654 for T_3). Which frequency band carries the information is therefore not stable across instruments, whose energy coverage also differs. Controlling for brightness with the log count rate as an extra input reduced the N1 LR gain to +0.069 [−0.013, +0.161]. The pooled RXTE+NICER test gave +0.145 [+0.028, +0.257], but it combines data already analysed and is not independent of the rows above it.

Summary. Before any test on unseen observations, every point estimate of the hard-like timing gain was positive. With LR, only the RXTE development interval excluded zero. With RF, every interval excluded zero, but RF was not yet a primary model.

4.4 Background robustness and tests on unseen observations

N2: unseen observations, LR fixed in advance. The primary LR gain on 143 hard-like N2 observations was +0.066 [−0.095, +0.180], which was not detected. The LR colour-only AUC was 0.660 and H+T-LR reached 0.726. At the frozen 0.5 threshold, H+T-LR had a *lower* observation-level balanced accuracy than H-LR (−0.080 [−0.166, −0.022]), indicating that its calibration did not carry over to the new observations. The RF gain on the same observations was +0.247 [+0.128, +0.384] (0.665 to 0.912; descriptive).

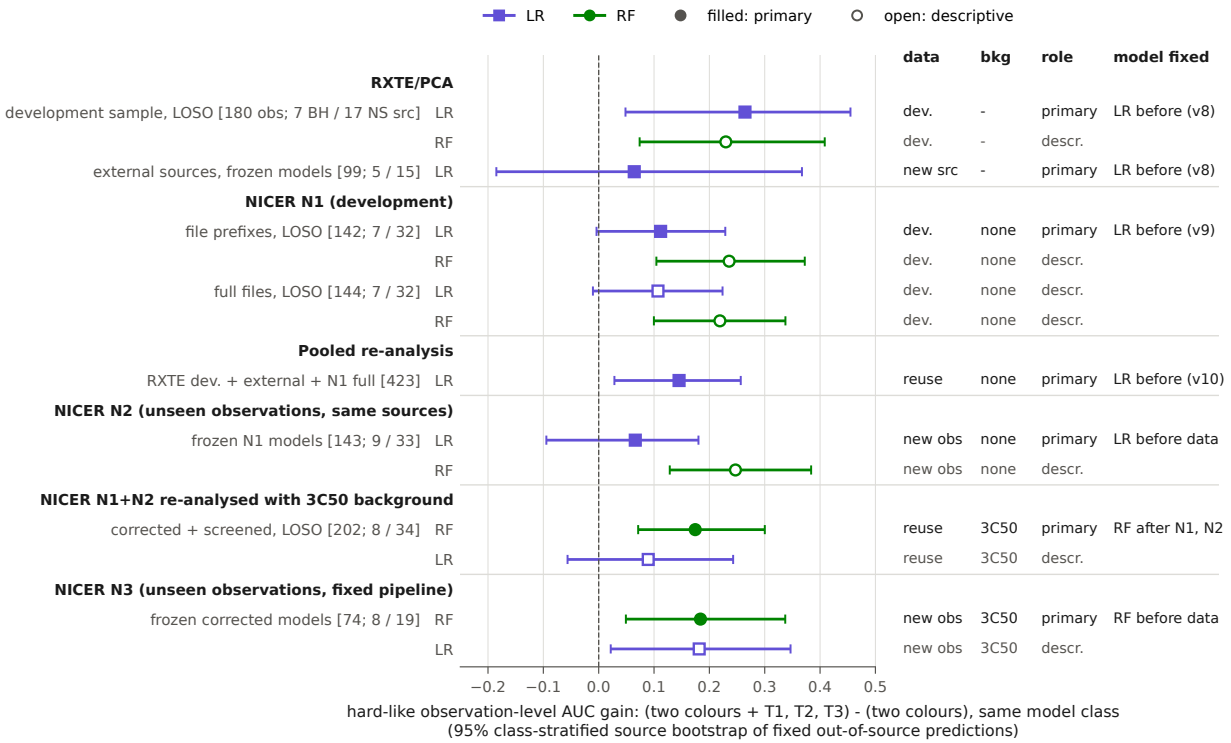


Figure 4: Hard-like observation-level AUC gain from adding fractional rms in 0.016–0.094, 0.10–1.0 and 1.0–4.0 Hz to two colours, for the same model class (LR violet squares; RF green circles). Filled: primary comparison of that round's plan; open: descriptive. Brackets give hard-like observations and BH/NS sources. "Data" marks development data used to form the hypothesis (dev.), sources never used in training (new src), observations never analysed before (new obs), or re-analysis of data already analysed (reuse). "Model fixed" gives when the primary model class was fixed relative to the data; all plans were written after earlier results. Rows sharing data are not independent: the pooled row combines the RXTE and N1 rows, and the 3C50 row re-analyses N1 and N2. Intervals: 95% class-stratified source bootstrap of fixed predictions, excluding training variability. Source: `results/final_timing_gain_across_versions.csv`; `results/v10a/v10a_pooled_gain.csv` for the N1 full-file LR row.

N1+N2 with 3C50 background correction. After correction and screening, the RF gain was +0.174 [+0.071, +0.300] (0.750 to 0.924; 202 hard-like observations of 8 BHs and 34 NSs). RF was made primary for this round after its gains had been seen in N1 and N2, and the data are re-used, so this result tests robustness to the background treatment. It is not an independent confirmation. The LR gain was +0.090 [−0.056, +0.243]. At source level, the RF AUC gain was +0.074 [−0.033, +0.206] and the balanced-accuracy gain +0.173 [+0.004, +0.357]. The background correction itself changed little: the median background fraction was below one per cent.

N3: unseen observations, fixed pipeline, RF fixed in advance. The background treatment, state thresholds, features and out-of-source RF models were all frozen before the N3 data were downloaded. The 74 hard-like N3 observations comprise 29 observations of 8 BHs and 45 of 19 NSs. Their composition is uneven: Cyg X-1 contributes 7 of the BH observations and 4U 1812−12 contributes 8 of the NS observations. The observation-level AUC rose from 0.680 [0.463, 0.870] for H-RF to 0.864 [0.714, 0.968] for H+T-RF, a gain of +0.184 [+0.049, +0.337] (primary, detected; Table 3, Figure 5). The descriptive checks were:

- LR showed a similar gain, +0.182 [+0.022, +0.347], from a colour-only AUC of 0.503;
- the soft-like RF gain was +0.005 [−0.015, +0.035];
- observation-level balanced accuracy at the frozen threshold rose by +0.137 [+0.019, +0.275];

Table 3: NICER N3 (unseen observations; frozen 3C50 pipeline; out-of-source models trained on corrected N1+N2). Point estimates with 95% class-stratified source-bootstrap intervals of fixed predictions. Observation-level rows use all sources of N3 in each draw; source-level rows and the observation balanced accuracy use a paired bootstrap over the 27 sources with hard-like observations.

Metric (hard-like unless stated)	Colours only	+ T_1–T_3	Difference	Role
Observation AUC, RF (74 obs)	0.680 [0.463, 0.870]	0.864 [0.714, 0.968]	+0.184 [+0.049, +0.337]	primary
Observation AUC, LR	0.503 [0.242, 0.766]	0.685 [0.375, 0.940]	+0.182 [+0.022, +0.347]	descr.
Obs. balanced accuracy, RF	0.607 [0.430, 0.769]	0.744 [0.557, 0.897]	+0.137 [+0.019, +0.275]	descr.
Source AUC, RF (27 src)	0.717 [0.507, 0.895]	0.842 [0.671, 0.961]	+0.125 [−0.053, +0.303]	descr.
Source balanced accuracy, RF	0.572 [0.421, 0.750]	0.671 [0.467, 0.859]	+0.099 [−0.053, +0.286]	descr.
Soft-like observation AUC, RF	0.938 [0.760, 1.000]	0.943 [0.772, 1.000]	+0.005 [−0.015, +0.035]	descr.
Obs. AUC, RF, 3C50-GTI timing	—	—	+0.190 [+0.065, +0.330]	descr.
Obs. AUC, RF, leave one BH out	—	—	+0.146 to +0.212 (8 values)	influence

- removing one BH source at a time left gains between +0.146 and +0.212;
- recomputing the timing features of 106 faint observations within the 3C50 GTIs gave +0.190 [+0.065, +0.330].

At source level (27 sources with hard-like observations), the AUC rose from 0.717 to 0.842, a gain of +0.125 [−0.053, +0.303], and the balanced-accuracy gain was +0.099 [−0.053, +0.286]. Neither was detected. The pre-specified test therefore confirms an observation-level gain on unseen observations. With equal weight per source the gain points in the same direction, but this sample cannot resolve it.

4.5 Secondary result: characteristic frequency at matched colour

We also asked whether, at matched colours, BHs have a lower characteristic variability frequency ν_c. Here ν_c is the power-weighted geometric-mean frequency of the 0.008–64 Hz (NICER) or 0.008–4 Hz (RXTE) cross-spectrum. The test proceeds in three steps:

1. residualise $\log_{10} \nu_c$ on the colours with a label-blind k-nearest-neighbour regression;
2. average the residuals within each source;
3. compare the BH and NS means by permuting source labels.

The details are in Appendix C. The BH-minus-NS difference D was negative in every NICER set and in RXTE hard-like data (−0.27 to −0.66 dex), but its significance varied:

- the pooled test gave $p = 0.0012$, but it re-uses data;
- in N2, $D = -0.66$ ($p = 0.022$) weakened to -0.46 ($p = 0.19$) when very hard observations, likely dominated by background, were removed post hoc;
- after 3C50 correction, N1+N2 gave $D = -0.51$ ($p = 0.073$);
- N3 gave $D = -0.43$ ($p = 0.21$) under the primary pipeline and $D = -0.30$ ($p = 0.41$) with timing restricted to the 3C50 GTIs;
- the RXTE external sources showed no difference ($D = -0.04$, $p = 0.78$).

We regard this as a consistent but unconfirmed lead. It is not evidence for mass scaling, because neither Eddington ratio nor mass was controlled.

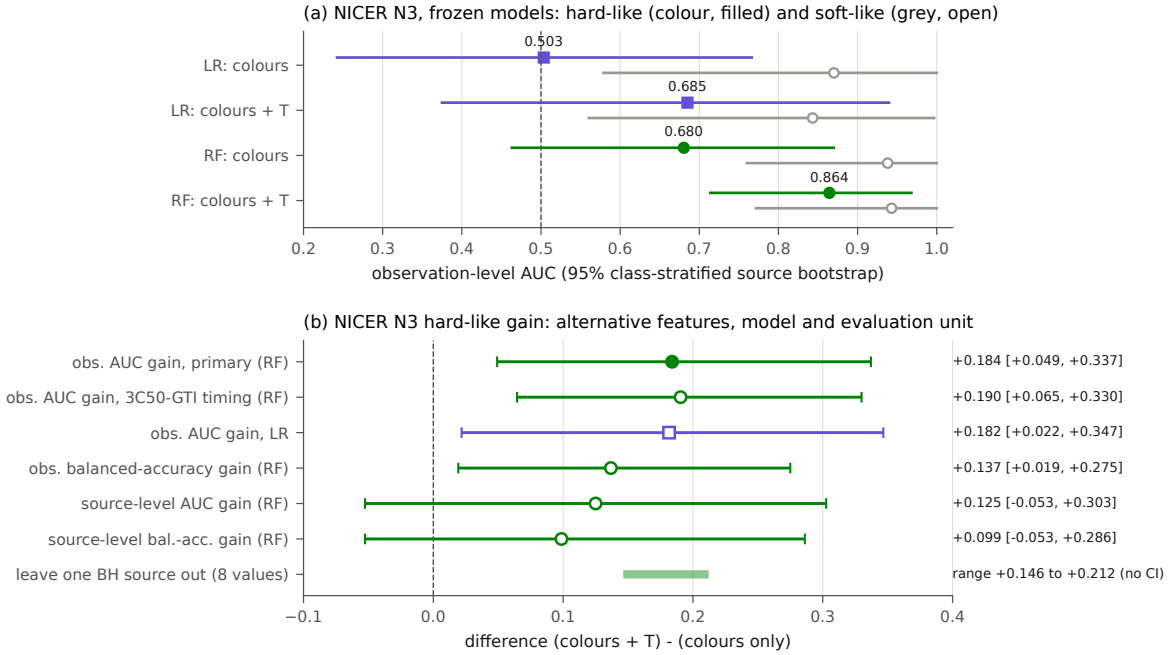


Figure 5: The N3 test on unseen NICER observations. (a) Observation-level AUC of frozen out-of-source models in hard-like (colour, filled; 74 observations, 8 BH/19 NS sources) and soft-like (grey, open; 120 observations) observations; numbers are hard-like point estimates. (b) Hard-like gain from adding T_1–T_3, under the primary specification (filled) and descriptive alternatives: timing features recomputed within the 3C50 good-time intervals for 106 faint observations; LR instead of RF; observation-level balanced accuracy at the frozen 0.5 threshold; source-level AUC and balanced accuracy (each source weighted equally); and the range of the primary gain when one BH source is removed at a time (bar; no interval). Note that rows use different metrics. Intervals: 95% class-stratified source bootstrap of fixed predictions. Source: `results/v13/v13a_gain.csv`, `v13a_gti_restricted.csv`, `v13a_influence.csv`.

5 Discussion

5.1 What has been learned beyond classification accuracy

Where the spectral signal lives. In the RXTE development sample, two hardness ratios reproduce the source-level performance of full-spectrum representations. Neither more bands, more observations nor more flexible models added detectable information. This connects the classification problem to the colour–colour phenomenology of Done and Gierliński (2003). It also suggests, without testing it, that spectral classifiers trained on similar PCA data (Pattnaik et al., 2021) may rely largely on the same two-colour information. We did not evaluate those models, so this remains a hypothesis.

Where it fails. The colour information is concentrated in soft-like observations. In hard-like observations, colour-only AUCs of 0.50–0.75 across five samples show that BH and NS colours overlap substantially. This quantifies, with intervals that respect source clustering, the state dependence noted by Pattnaik et al. (2021). It is also consistent with Burke et al. (2017), whose hard-state differences lie in Comptonization parameters that 2–25 keV count colours sample only weakly.

What variability adds. Within hard-like observations, adding three low-frequency rms amplitudes to the colours increased the observation-level AUC (point estimate) in every sample analysed. As a primary test, the increase was detected in the RXTE development sample, in the N1+N2 3C50 re-analysis, and in the N3 test with a pipeline and model fixed before download; it was not detected in the RXTE external set, in N1 with LR, or in N2 with LR. The gain was absent in soft-like observations, where colours already separate the classes, so timing complements colours

rather than replacing them. This is not the high-frequency criterion of Sunyaev and Revnivtsev (2000). Our exploratory event-mode analysis found 512–1024 Hz power detectable in only 1 of 17 NSs in single RXTE observations, and its implementation check failed (Appendix B). Nor does the result contradict the broad similarity of BH and NS power colours (Heil et al., 2015; Gardenier and Uttley, 2018): tracks that are similar overall can still differ in amplitude at a given colour.

How robust the gain is to modelling choices. RF gains were consistently +0.17 to +0.25. LR gains ranged from +0.07 to +0.26, and LR calibration failed to transfer to N2. This points to a non-linear or threshold-like relation between colours, rms and class, but it is not a mechanism. The band that matters also differed between instruments (T_1 for RXTE, T_2 and T_3 for NICER), and the two instruments sample different energies. We therefore do not attribute the gain to a particular power-spectral component.

5.2 What remains unproven

1. *Generalisation to new sources.* N2 and N3 are new observations of sources already in the sample. The out-of-source models never trained on the tested source, but the source population, and the hypotheses, came from the same sources. Observations of one outburst are correlated. In N3 the source-level gain (+0.125) was not detected. Confirmation needs BHs not yet used, such as future transients.

2. *One confirmatory test of the RF specification.* RF became the primary model only after its gains had been seen. N3 is the only test in which it was fixed before download. The N1+N2 re-analysis and the pooled test re-use data and are not replications.

3. *Selection on a timing quantity.* Hard-like observations are selected by $r > 0.10$, a variability amplitude that overlaps T_2 and T_3. Labels were not used, and both models are evaluated on the same subset, so this selection does not by itself create a spurious gain. It does mean that the result is conditional on an rms-defined subset, not on a spectrally defined hard state, and that a practical classifier needs timing to identify the stratum. A natural question is whether a single total rms adds as much as the three bands. We did not compare colours plus total rms with colours plus T_1–T_3, so we cannot say whether band structure or amplitude alone carries the information.

4. *Weighting and dependence.* Observation-level AUCs weight sources by their number of observations. Source-level metrics give equal weight but have little power with 8 BHs. We did not run a source-equal-weight observation-level analysis, nor one blocked by outburst.

5. *Background and selection effects.* The 3C50 failures (mainly early gain-calibration files) are not random. Background time variability is not modelled. The mismatch between 3C50 and timing GTIs was checked only for N3.

6. *Physical confounders.* Eddington ratio, inclination and NS subclass (atoll, Z source, accreting millisecond pulsar) were not controlled; only a count-rate proxy for brightness was tried.

7. *Uncertainty.* The bootstrap intervals include source sampling only. They exclude refitting variability, measurement error in the features and the labels, and the multiplicity of comparisons made over the project.

5.3 How this extends prior work

Several elements are not new: evaluation with sources held out (Pattnaik et al., 2021; de Beurs et al., 2022), the observation that hard and intermediate states are harder to classify (Pattnaik

et al., 2021), and the use of timing to separate BHs from NSs (Sunyaev and Revnivtsev, 2000; Wijnands and van der Klis, 1999; Klein-Wolt and van der Klis, 2008). What this work adds is narrower:

- an ablation showing that two colours carry the detectable PCA spectral information in this sample;
- an estimate of the state dependence with source-level uncertainty;
- a specific test of whether low-frequency variability adds information beyond colours within hard-like observations of held-out sources, repeated across two instruments and a background re-analysis, and confirmed once on unseen observations with a frozen pipeline.

The observation-split evaluation of Garg et al. (2026) answers a different question, namely recognition of sources already seen in training. In our data, splitting by observation inflated balanced accuracy by 0.04–0.16, but we did not evaluate their model, and we do not claim that it fails on new sources. Their "variance of the data" is a spectral statistic, not a Fourier timing feature, so the two results are not directly comparable. For MAXI colours (Appendix B), we reproduced the published results of de Beurs et al. (2022). With strictly dynamical BH labels, linear models on MAXI colours had little discriminating power, while non-linear models reached a source-level AUC near 0.8. The samples, labels and metrics differ, so these numbers do not rank the methods. No result here supports claims of being first, of state-of-the-art performance, or of outperforming earlier work.

5.4 Next steps

The most informative next step is to apply the frozen N3 pipeline and RF model, without changes, to BH and NS sources never used here, for example new transients observed by NICER. Before that, analyses that could change the interpretation can be done with existing data:

- colours plus total rms against colours plus T_1–T_3;
- source-equal-weight and outburst-blocked versions of the hard-like gain;
- matching the 3C50 and timing GTIs in all NICER sets;
- a fair comparison with published models on identical data and splits.

6 Conclusions

1. For RXTE/PCA 5–25 keV spectra of 31 held-out sources, two hardness ratios gave a source-level AUC of 0.923 [0.804, 0.994]. No other spectral representation, energy band, sample size or model search tried here added detectable information. This is not evidence that the colours contain all of the information.
2. Colour information is state dependent. The observation-level AUC was 0.954 for soft-like and 0.646 for hard-like RXTE observations (difference +0.309 [+0.099, +0.510]). Every NICER set showed the same ordering.
3. Within rms-defined hard-like observations, adding 0.016–4 Hz rms amplitudes to the colours increased the observation-level AUC in every sample. On 198 unseen NICER observations with a frozen pipeline and a random forest fixed before download, the hard-like AUC rose from 0.680 to 0.864 (+0.184 [+0.049, +0.337]). The source-level gain (+0.125 [−0.053, +0.303]) was not detected.

4. All tests on new data used the same sources. Confirmation on new BHs is the decisive missing step.

5. A lower characteristic variability frequency of BHs at matched colour is a consistent but unconfirmed secondary lead. It does not support any claim about mass scaling, event horizons or new physics.

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Data and code availability

All analysis code, configuration files, time-stamped analysis plans, decision logs and result tables are in the repository `https://github.com/Chipwwwwww/XRB-pilot` **[confirm public visibility and archive with a DOI]**. This draft uses commit `666c2e0`. Raw RXTE and NICER data are public at HEASARC. Download URLs, byte counts and SHA-256 checksums are logged in `logs/downloads.jsonl`. Figures 2–5 and C.1 were redrawn from committed result files by `paper_draft/scripts/make_figures.py`, without re-running any analysis.

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A Correspondence with the repository analysis rounds

The repository records the work as numbered rounds (v1–v14), each with its own plan, script set and result directory. Table A.1 maps them to the sections of this paper. The rounds are listed for traceability only; the paper is organised by scientific question.

Table A.1: Repository rounds used in this paper. "Plan" files are `reports/preregistration_vN.md`; results are under `results/vN*/`.

Round	Content used here	Section
v1–v2	RXTE pilot (8 sources) and development sample (31 sources), LOSO, representations A, B, H, HI	2.2, 4.1
v3	nested increment H+B−H, 3–25 keV band, BH candidates (descriptive)	4.1, B
v4	algorithm search, burst check, HEXTE, earlier gain epochs, all observations	4.1, B
v5–v6	Standard-1 timing features, external sources, permutation test	3.2, C
v7	state classes, MINBAR burst exclusion, persistent BHs, MAXI	3.3, 4.2, B
v8	hard-like timing gain (RXTE), external hard-like test, event-mode high frequencies, MAXI N_{H}	4.3, B
v9–v10	NICER N1, pooled RXTE+NICER	2.3, 4.3
v11	NICER N2 (unseen observations, frozen models)	4.4
v12	3C50 background re-analysis of N1+N2	2.4, 4.4
v13	NICER N3 (unseen observations, frozen pipeline, RF fixed)	4.4
v14	cross-round summary (no new analysis)	—

B Additional results

The following results support the main text but are not part of its argument. All are taken from committed result files.

- *Burst contamination (v4, v7).* Matching the 456 development observations to MINBAR within the PCA field of view flagged 58 observations, all NSs. Excluding them did not change the colour-baseline or full-spectrum conclusions.
- *BH candidates (v3c).* Fourteen BlackCAT candidates were scored with models trained on confirmed sources. This is descriptive only, and the candidates were never used as labels.
- *External sources, all states (v6).* Frozen colour models reached a source-level AUC of 1.000 with and without timing on 23 external sources. Source-level balanced accuracy rose from 0.868 to 0.974 (+0.105 [0.000, +0.237]), which was not detected.
- *MAXI (v7, v8, v9).* The published MAXI results of de Beurs et al. (2022) were reproduced from their public data. With dynamically confirmed BHs only (13 BHs and 48 non-pulsing NSs), LR on MAXI colours gave source-level AUCs of 0.56–0.59. Neither primary comparison was detected. k-nearest-neighbour models on two colours reached 0.80. Adding a colour that includes the 2–4 keV band to a single ≥ 4 keV colour still helped after correcting for Galactic N_{H} (RF +0.377 [+0.160, +0.595]). However, the corrected single-colour comparison model had a source-level AUC of only 0.361, so part of this gain reflects a baseline below chance.
- *High-frequency timing (v8).* RXTE event-mode cross-spectra (4–1024 Hz) failed a pre-specified implementation check (Spearman correlation 0.81–0.82 with Standard-1 features, against the required 0.90), so all results are exploratory. After a post hoc subtraction of a white component common to the PCUs, 512–1024 Hz power was detected above 3σ in 1 of 17 NSs and in none of 6 BHs. High-frequency features added little beyond the low-frequency ones (RF +0.009).

C Characteristic frequency: definition and full results

For NICER, $\nu_{\mathrm{c}} = \exp(\sum_b w_b \ln \nu_b / \sum_b w_b)$ over 13 octave bins between 1/128 and 64 Hz, with $w_b = \max(P_b, 0)$ the band cross-power. ν_{c} is missing when the total rms in these bins is below

0.05. The RXTE definition (9 bins, 1/128–4 Hz) divides w_b by the logarithmic bin width; the two definitions are therefore not identical. The test proceeds as follows:

1. residualise $\log_{10}\nu_c$ on standardised colours with a $k = 25$ nearest-neighbour regression that excludes the observation's own source and ignores labels;
2. average the residuals per source;
3. take $D = \text{mean(BH)} - \text{mean(NS)}$;
4. permute source labels 20 000 times; the p-value is two-sided.

Figure C.1 lists all primary, descriptive and post hoc results. Frozen thresholds ν^* were also defined before N2 and N3 (1.30 and 1.336 Hz; a hard-like observation with $\nu_c < \nu^*$ is classified as a BH). In N3 this rule gave observation-level sensitivity 0.79 and specificity 0.76, and at source level classified 7 of 8 BHs and 13 of 19 NSs correctly.

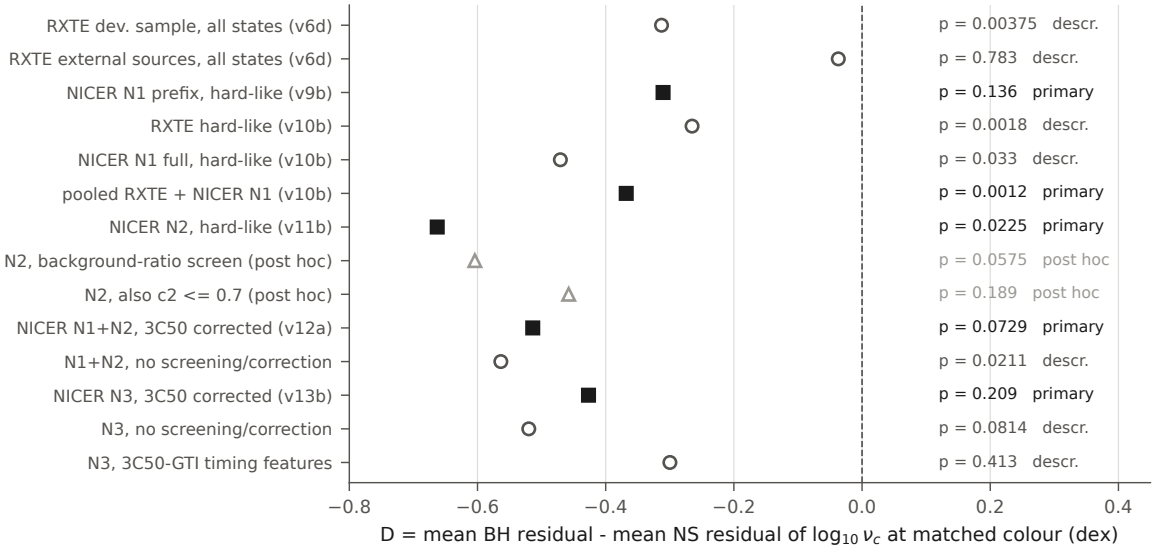


Figure C.1: Colour-conditional difference in $\log_{10}\nu_c$ between BH and NS sources (negative: BHs lower at matched colour). Squares: primary tests; circles: descriptive; triangles: post hoc. p-values from 20 000 source-label permutations. Rows share data (the pooled and 3C50 rows re-use earlier sets), so the p-values are not independent, and no multiplicity correction is applied. Source: `results/final_nuc_difference_across_versions.csv`, `results/v11/v11b_post_hoc_screens.csv`, `results/v12/v12a_nuc.csv`, `results/v13/v13b_nuc.csv`.

D Literature comparison

Table D.1 contrasts the designs; it is not a ranking of performance. The bibliographic fields were verified against search-engine records of arXiv, ADS and publisher pages on 2026 October 9. The full texts could not be retrieved from the writing environment, so method details for Garg et al. (2026) (observation-wise 80:20 split) and the state remark in Pattnaik et al. (2021) come from the project's earlier reading of those papers and should be re-checked ([**verify against full text**]).

Table D.1: Design comparison with related machine-learning work. Numbers are not comparable across rows.

	Pattnaik et al. (2021)	de Beurs et al. (2022)	Garg et al. (2026)	This work
Data	RXTE/PCA 5–25 keV, 43 channels; 61 LMXBs	MAXI colour–colour–intensity	RXTE spectra of the Pattnaik sample (+ errors, fit parameters)	RXTE/PCA colours and spectra; NICER colours; 0.016–4 Hz rms
Labels	BHs incl. candidates[†]	BH (incl. candidates), non-pulsing NS, pulsar	as Pattnaik, one ObsID set corrected	dynamical BHs only; bursters and pulsars
Split	sources held out	sources held out	observations (80:20)[†]	sources held out; unseen observations with frozen models
Metric	mean source accuracy $87 \pm 13\%$	accuracy	accuracy 90–94%	observation- and source-level AUC and balanced accuracy, source bootstrap
State resolution	misclassification noted in hard/intermediate states[†]	—	—	rms-defined classes, primary test of difference

[†]From the project's reading of the paper; to be re-checked against the full text.